Score Integrity Under Missing Modalities: A Renormalizing Multimodal Scoring Architecture for Automated Viva Assessment

B. K. H. M. B. Lakshan Herath
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
buddhilaherath28@gmail.com

L. Supun Buddhika Hemarathne
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
liyanagesupun10@gmail.com

Akila Manujith Senarathna
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
akilamanujith@gmail.com

H. Induwara Bandara Wickramarathne
Dept. of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
haritheb@gmail.com

Abstract*—Viva voce examinations assess reasoning and communication under direct questioning, but their scoring is hard to standardize because examiners must integrate verbal delivery, vocal behaviour and non-verbal cues. Automating any part of this raises a problem usually left implicit: what a scoring function should do when a signal cannot be measured. Treating an unmeasured signal as zero converts an acquisition failure into an academic penalty. This paper presents Stage-1, a deterministic multimodal scoring architecture for recorded viva sessions built on two commitments: missing measurements are omitted and the remaining weights renormalized, and an enforced exclusion boundary prevents unvalidated, diagnostic and language-model-derived signals from reaching the official score. Three feature families are fused: visual engagement, acoustic delivery quality and transcript fluency. We evaluate the scoring function through a twenty-thousand-sample simulation over its input space, driven by the production scoring code itself. Renormalization reduces the mean absolute score disturbance caused by a missing signal from nine point one two to three point zero four points on a hundred-point scale, a threefold improvement over zero-imputation; it removes a systematic downward bias reaching twenty-two points when a whole modality is lost, and more than halves the proportion of candidates whose letter grade is disturbed. The advantage widens when degradation is correlated across signals, the realistic equipment-failure mode. Perturbing any single input by a tenth moves the score by at most one and a half points, and sweeping the unfitted constants leaves candidate ordering essentially unchanged. Invariance tests confirm that language-model criterion scores and diagnostic aggregates cannot alter the official score. No human-examiner ground truth exists yet; we therefore claim no agreement with human grades and specify the validation study required to establish it.

Keywords—automated assessment, multimodal fusion, viva voce, missing-data robustness, educational technology, speech analysis, score integrity

I. Introduction

Viva voce assessment occupies an important position in higher education because it allows an examiner to probe reasoning, ask follow-up questions and observe communication under direct interaction. The same characteristics make it demanding to administer consistently: subjective judgment is affected by contextual factors and fatigue, particularly when many open-ended responses must be judged in sequence [1], and there is growing interest in estimating confidence and other learner states from speech and behavioural signals [2], [3].

Automating part of this process is attractive, but a numerical score can appear objective while remaining difficult to audit. Multimodal machine learning involves representation, alignment, fusion and co-learning challenges [4], and deployment becomes harder when one or more modalities are unavailable [5]. An assessment system must therefore answer two questions usually treated as implementation details: which observable signals may contribute to a grade, and what happens when a permitted signal cannot be measured. Both have consequences for students. If a microphone clips or a voice-quality estimator fails to converge, an implementation that substitutes zero silently penalises the candidate for a property of the recording equipment. Conversely, if every computable signal may influence the grade because it is available, a classifier trained on an unrelated domain can move a mark without anyone having validated its relevance.

This paper studies the VivaEvaluationEngine, the viva-scoring component of the Gradex academic-assessment platform, restricting attention to its production scoring core, Stage-1 v1.1, which fuses engagement, acoustic delivery and transcript fluency. The architecture separates official scoring inputs from diagnostic and experimental signals, which matters for systems in which large language models (LLMs) supply qualitative feedback that should not contaminate an independent behavioural score.

The paper makes three contributions. First, it formalizes a three-family scoring architecture with bounded normalization and dynamic renormalization over missing sub-features and missing families, under an explicit policy distinguishing *missing* from *poor*. Second, it introduces an enforced exclusion boundary and feature-validation ledger, in which a signal must earn the right to grade through validation rather than availability, asserted by invariance tests rather than by convention. Third, it quantifies the scoring function through a simulation over 20,000 sampled feature vectors: a missing-data ablation against a zero-imputation baseline, a

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

perturbation sensitivity analysis, a sweep of every unfitted constant, and a replication under correlated degradation. To our knowledge the cost of the zero-imputation shortcut has not previously been quantified in an educational-assessment context. We also state rather than obscure the limits of the present evidence: no human-examiner ground truth exists, so no agreement statistic is claimed, and Section X specifies the study required to obtain one.

II. Related Work

A. Multimodal Fusion and Missing Modalities

Multimodal machine learning identifies fusion, representation, alignment and co-learning as core challenges [4]. Where a modality can be absent at inference time, missing-modality robustness becomes an explicit architectural concern; Wang et al. evaluated a tri-modal fusion architecture for medical data that addresses missing modalities during inference [5]. That work learns to compensate for absent modalities, whereas the present work applies a simpler deterministic principle to viva scoring, omitting a missing signal rather than reconstructing it, because in assessment a learned imputation would itself require validation before it could defensibly influence a grade.

B. Behavioural Signals in Oral Assessment

Speech analysis supplies prosodic and voice-quality measurements. Reddy et al. used audio analysis to study student confidence in viva responses [3]; Khalil et al. reviewed deep-learning speech emotion recognition [6]; and Tao et al. estimated confidence from behavioural signals in immersive learning environments [2]. On the visual side this implementation uses MediaPipe face-landmark processing [7], while prior work has studied micro-expression analysis extensively [8]. Emotional state and academic performance are distinct constructs, so emotion, gaze, head pose and blink measures are treated as separate signals with separate validation status. Transcription uses Whisper [9]; BERT-style models show the potential of semantic modelling for future content components [10], and earlier interview systems used NLP to evaluate answers [11]. These studies motivate behavioural signals as evidence but do not define how a production score should combine them under partial observability.

C. Automated Viva Systems and Explainability

Cao and Zahid proposed AutoViva, generating personalized viva questions for coursework authentication [12]; Ipeirotis and Rizakos described a voice-AI system in which an agent conducts examinations and a panel of LLMs contributes to transcript grading [13]. Both expand what an automated examiner can do; this paper addresses how the scoring layer beneath such components should be engineered so its inputs remain bounded, auditable and stable when signals go missing. Interpretability methods such as LIME [14] and SHAP [15] explain decisions after prediction; this implementation provides no formal attribution layer, and its contribution to auditability is structural, since every official input belongs to a named family with a recorded validation status. We treat this as score provenance, the weaker claim.

III. System Architecture

The engine supports two operational contexts: assessment of an uploaded recording, and a live examiner copilot suggesting follow-up questions. The copilot is advisory and human-in-the-loop, never addressing the student and contributing nothing to the mark. Both converge on the same file-based scoring path once recording ends. One canonical scoring path rather than separate live and offline models removes a class of defect in which a mark would depend on the mode of examination.

The video branch samples frames with OpenCV, detects faces, applies dense landmarks with the MediaPipe Tasks FaceLandmarker and derives engagement, gaze, blink and pose measures; a custom EfficientNet-B0 emotion model exists in the wider pipeline but emotion labels are not part of Stage-1. Three integrity gates operate upstream of scoring: face quality, a multi-face gate detecting a sustained secondary face, and a lip-motion gate requiring mouth motion proportional to transcribed speech, failure of the latter two marking the assessment INCOMPLETE rather than producing a degraded number. The audio branch extracts the soundtrack, diarizes to isolate the student track and transcribes with Whisper [9]; of the stored pitch statistics, RMS energy, MFCCs, jitter, shimmer and harmonics-to-noise ratio (HNR), only pitch variability, HNR and jitter/shimmer articulation enter Stage-1.

Three LLM-oriented services are architecturally separate from the scorer: conversation understanding segments the dialogue, question-answer relevance classifies whether an answer addresses the question, and a technical-accuracy service scores transcript coverage against an examiner-supplied rubric. Each is invoked explicitly and none is imported by the scoring path.

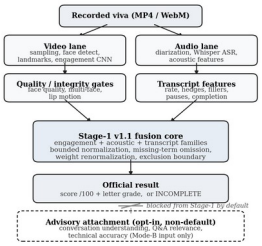


Fig. 1. Canonical viva analysis path. Uploaded and live recordings converge on one Stage-1 scoring path; advisory LLM services sit outside the scoring boundary.

IV. Feature Extraction and Normalization

Every scored quantity is mapped into [0, 1] by a bounded, monotone and deterministic function: boundedness ensures no single input dominates, monotonicity makes each input's direction inspectable, and determinism ensures a recording yields the same mark on re-analysis. Let *clamp*(x, a, b) denote truncation of x to [a, b].

A. Engagement Family

The engagement family contains the mean CNN engagement estimate, e_cnn in [0, 1], and normalized facial confidence, c_f = confidence_score/100. When both are present,

$$E_{\text{eng}} = \frac{e_{\text{cnn}} + c_f}{2} \tag{1}$$

If only one is available, that value forms the family score, so a missing confidence estimate does not become a zero-confidence student.

B. Acoustic Family

The acoustic family contains three normalized quantities. Pitch stability and HNR clarity are

$$s_p = 1 - \text{clamp}\left(\frac{\sigma_f}{\theta_p},\ 0,\ 1\right) \tag{2}$$

$$s_{\mathrm{hnr}} = \mathrm{clamp}\Big(\frac{\mathrm{HNR}}{\theta_h},\, 0,\, 1\Big) \tag{3}$$

where sigma_f is pitch standard deviation in hertz, theta_p = 120 Hz and theta_h = 30 dB. For a lower-is-better voice-quality metric x, the implemented quality map is the clamped linear interpolation

$$q(x) = \mathrm{clamp}\Big(\frac{x_{\mathrm{poor}} - x}{x_{\mathrm{poor}} - x_{\mathrm{good}}},\, 0,\, 1\Big) \tag{4}$$

so that articulation is

$$s_a = \tfrac{1}{2} q(\mathrm{jitter}) + \tfrac{1}{2} q(\mathrm{shimmer}) \tag{5}$$

The configured thresholds are jitter good/poor = 0.005/0.030 and shimmer good/poor = 0.030/0.150, engineering choices rather than fitted parameters; their influence is quantified in Section VII-C. In the implementation HNR clarity and articulation are gated on a single voice-quality flag, so a failure of that estimator removes both terms together; the ablation treats them separately to isolate each contribution.

C. Transcript Family

The transcript family uses five deterministic measures needing no model answer. Speech rate is scored by a band function,

$$s_r = 0.65 + 0.35 \cdot 1[\,\rho_{\mathrm{lo}} \le \mathrm{WPM} \le \rho_{\mathrm{hi}}\,] \tag{6}$$

with rho_lo = 120 and rho_hi = 160 words per minute and an out-of-band value of 0.65. Hedge, filler and long-pause scores are

$$s_{\mathrm{hed}} = 1 - \min\Big(1, \tfrac{N_h}{8}\Big), \quad s_{\mathrm{fil}} = 1 - \min\Big(1, \tfrac{N_f}{12}\Big) \tag{7}$$

$$s_\ell = 1 - \min\Big(1, \frac{N_\ell}{6}\Big) \tag{8}$$

where N_h, N_f and N_l are hedge, filler and greater-than-two-second pause counts. Sentence completion is clamped to [0, 1]. As with the acoustic thresholds, the band limits and divisors are prototype configuration values rather than fitted parameters.

D. The Exclusion Boundary

The engine computes far more measurements than it grades on, because computing a feature is not the same as licensing it to grade. LLM criterion scores, diagnostic engagement blends, interface sliders, legacy score fields, word-count heuristics, RMS loudness, speech-emotion labels, MFCCs, raw gaze coordinates and raw blink counts are excluded from Stage-1. Loudness is instructive: trivially computable and intuitively appealing, but louder is not better and no validated mapping from recording energy to viva quality exists.

The boundary is enforced at three levels: a versioned canonical feature-vector schema admits only permitted fields; the scoring functions read from that schema rather than raw analysis output; and two invariance tests assert that neither LLM evaluation scores nor diagnostic aggregates change the computed score. A ledger records, for every signal, whether it is measured, normalized, scored, diagnostic or awaiting validation. We regard the fifth row of Table I as a design outcome rather than unfinished work: five computable signals are held out because the evidence to justify including them does not exist.

TABLE I. STAGE-1 SCORING INPUTS AND THE EXCLUSION BOUNDARY

Family / status	Inputs	Policy
Engagement	CNN mean engagement; facial confidence	Scored; equal weight
Acoustic	pitch stability; HNR clarity; jitter/shimmer articulation	Scored; missing terms omitted
Transcript	rate band; hedges; fillers; long pauses; completion	Scored; missing terms omitted
Diagnostic	emotion, gaze coordinates, blink count, RMS, MFCC, SER	Stored as evidence; never Stage-1
Held out pending validation	CNN-LSTM temporal score; response structure; pause totals; head-event detection	Computable; excluded until validated
Advisory / optional	LLM narrative, Q&A relevance, technical accuracy	Outside Stage-1 unless fused via Mode B

V. STAGE-1 SCORING AND THE MISSING-DATA POLICY

Let A_i denote the set of available sub-features of family i. The family score is the mean of its available sub-features,

$$E_i = \frac{1}{|A_i|} \sum_{k \in A_i} s_k, \qquad A_i \neq \emptyset \tag{9}$$

and a family with no available sub-feature is itself unavailable. Let F be the set of available families, with equal nominal weights w_eng = w_ac = w_tr = 1. The official AI-performance score is

$$S_{\mathrm{AI}} = 100 \cdot \sum_{i \in F} \tilde{w}_i E_i, \qquad \tilde{w}_i = \frac{w_i}{\sum_{j \in F} w_j} \tag{10}$$

The renormalization in (10) states the design principle formally: weight belonging to an unmeasurable signal is redistributed across the signals that were measured, rather than spent on a fabricated value. The alternative, adopted as the comparison baseline in Section VII, is zero-imputation, which retains the full nominal weight set while every unavailable sub-feature contributes zero:

$$S^0_{\mathrm{AI}} = 100 \cdot \sum_{i \in \mathcal{F}} \frac{w_i}{\sum_{j \in \mathcal{F}} w_j} E^0_i, \qquad E^0_i = \frac{1}{|K_i|} \sum_{k \in K_i} \hat{s}_k \tag{11}$$

where K_i is the nominal sub-feature set of family i and s-hat_k = s_k when k is available and 0 otherwise. This is what an implementation produces when missing values are initialized to zero.

The policy is therefore specified by four rules. An unavailable sub-feature is omitted and the remainder reweighted equally; an unavailable family is omitted and the remaining weights renormalized; a failed visible-face or lip-motion gate returns INCOMPLETE; and if no scorable family remains, no numerical score is produced. Refusing to score is a legitimate output.

Two final modes are supported. Mode A reports the behavioural score alone; Mode B fuses a separately supplied technical-accuracy score T in [0, 10] with equal weight:

$$S_{\mathrm{final}} = S_{\mathrm{AI}} \quad (\mathrm{Mode\ A}) \tag{12}$$

$$S_{\mathrm{final}} = 0.5\, S_{\mathrm{AI}} + 0.5\big(\tfrac{T}{10} \times 100\big) \quad (\mathrm{Mode\ B}) \tag{13}$$

Mode B remains pending until T is supplied; a missing technical score is never read as zero, which would halve a candidate's mark because a rubric had not been uploaded. Grades map to A+ for 90-100, A for 80-89, B+ for 70-79, B for 60-69, C+ for 50-59, C for 40-49 and F below 40. As a worked example, with pitch stability 0.80, HNR unavailable, articulation 0.60 and engagement 0.75, and the transcript family unavailable, the two remaining families are equally weighted and the score is 72.5; under zero-imputation the same recording scores 41.7, nearly three grade bands lower. A locked fixture pins the behaviour: a fixed all-families vector must yield 73.76 (B+, VALID) on every run.

VI. Experimental Methodology

A. Scope and Rationale

The claims here concern the behaviour of a scoring function, not its agreement with human examiners. No human-examiner ground-truth dataset exists for this system, so no correlation, accuracy, agreement or kappa statistic against human grades is reported or implied anywhere in this paper; the available recordings are engineering fixtures, not a representative corpus, and Section X specifies the study needed to close this gap. The restriction does not preclude quantitative evaluation: Stage-1 is a closed-form deterministic function of a known input space, so its robustness, sensitivity and dependence on unfitted constants can be measured exhaustively without labels.

B. Simulation Design

We sampled N = 20,000 independent feature vectors uniformly across plausible operating ranges: e_cnn and c_f in [0.30, 0.95]; pitch standard deviation in [10, 140] Hz; HNR in [5, 32] dB; jitter in [0.002, 0.040]; shimmer in [0.020, 0.180]; speech rate in [80, 200] WPM; completion in [0.50, 1.00]. Hedge, filler and long-pause counts came from Poisson distributions with means 3, 5 and 2. Ranges span the non-saturated region of each normalizer.

Every reported figure is produced by driving the production scoring function itself rather than a reimplementation of the equations above, so the results describe the deployed artefact. The generator is seeded and the script distributed with the paper, making Section VII reproducible. The baseline population has mean 59.28 and standard deviation 7.36, characterizing the sampler, not a cohort.

C. Experiments

Experiment 1, missing-data ablation. For each of thirteen conditions, covering every individual sub-feature and each whole family, we recomputed every sampled vector under both policies (10) and (11), recording the signed and absolute score change, the proportion of samples whose letter grade changed, and the rank correlation with the unablated score. *Experiment 2, perturbation sensitivity.* Each raw input was scaled by +/-10% in turn with all others fixed, quantifying each signal's leverage. *Experiment 3, threshold sensitivity.* Each unfitted constant, and the speech-rate band limits, was swept across a plausible range with the others held at configured values. *Experiment 4, correlated degradation.* Experiment 1 was repeated on a second population generated from two latent factors, recording quality and speaker fluency, so that a single degraded microphone or camera moves several signals together.

Experiment 5, invariance and behaviour tests. The engine suite comprises 189 passing test functions across 16 files, re-verified on 1 September 2026 under Python 3.10.11; fifteen target the scoring core and 174 cover surrounding components, which are evidence about upstream behaviour rather than the scoring formula.

VII. Results

A. Renormalization Versus Zero-Imputation

Renormalization reduces the mean absolute score disturbance caused by a missing signal from 9.12 points under zero-imputation to 3.04, a factor of 3.00 pooled across thirteen conditions. The signed bias differs more sharply: zero-imputation is punitive in every condition, with mean shifts from -3.88 to -22.61 points, whereas renormalization stays between -2.39 and +1.51 for individual sub-features.

The whole-family conditions show the stakes. Losing the transcript family, which a failed transcription would cause, shifts the mean score by -22.61 points under zero-imputation against -4.27 under renormalization; losing the engagement family costs -20.79 against -1.55. Under zero-imputation a microphone or camera fault is worth roughly two grade bands. Averaged across conditions, 65.8% of candidates change letter grade under zero-imputation against 29.5% under renormalization.

TABLE II. Missing-Data Ablation: Renormalization vs Zero-Imputation

Ablated signal	Renorm. mean ΔS	Zero mean ΔS	Renorm. grade chg	Zero grade chg
CNN engagement	-0.02	-10.41	35.6%	88.4%
Facial confidence	+0.02	-10.38	36.2%	88.3%
Pitch stability	+1.51	-4.28	34.4%	43.0%
HNR clarity	-2.39	-6.88	34.8%	67.6%
Articulation	+0.88	-4.70	30.0%	46.9%
Transcript sub-features (range)	-0.72 to +0.80	-5.09 to -3.88	10.7-13.7%	38.8-51.0%
Engagement family	-1.55	-20.79	42.7%	99.7%
Acoustic family	+5.84	-15.87	60.0%	96.8%
Transcript family	-4.27	-22.61	48.3%	99.7%
Pooled mean absolute ΔS	3.04	9.12	29.5%	65.8%

One result runs against the design and we report it rather than omit it. Averaged over conditions, zero-imputation preserves candidate *rank* marginally better (mean Spearman rho 0.911 against 0.896), because subtracting a fixed weighted term is close to a monotone transformation, whereas redistributing weight alters the relative contribution of the surviving signals. Zero-imputation preserves ordering while destroying calibration; renormalization preserves calibration at a small ordering cost. For criterion-referenced assessment against fixed grade boundaries calibration determines the mark, and we accept the trade; a norm-referenced setting would weigh this differently.

The acoustic family is the only condition producing a positive mean shift (+5.84): under the sampled ranges its sub-features score lower than the other families, so removing them raises the renormalized mean. Renormalization is unbiased with respect to *missingness as evidence*, not the score level.

B. Sensitivity to Individual Inputs

No single input dominates: a 10% change in any raw input moves the score by at most 1.52 points and on average by no more than 1.03. CNN engagement and facial confidence lead at 1.030 and 1.027 mean absolute change, HNR and speech rate follow at 0.606 and 0.548, and the remaining six inputs

fall between 0.215 and 0.516. The ordering follows from family structure rather than tuning: the engagement family has two sub-features carrying one sixth of total weight each, while each of the five transcript sub-features carries one fifteenth, so engagement inputs have roughly four times the leverage of a transcript input.

This asymmetry is a genuine finding about the current configuration and not an intended design property: the two least externally-validated inputs, CNN engagement and facial confidence, are also the two with the greatest influence on the mark, visible only because the sensitivity analysis was run. We regard it as the most important target for the validation study in Section X. Speech rate shows the largest 95th-percentile movement relative to its mean, as expected of the only step-valued normalizer, arguing for a smooth replacement once labelled data indicate where the transition lies.

C. Sensitivity to Unfitted Constants

The most direct objection to this design is that its constants are chosen rather than fitted. The sweep does not dismiss that objection but bounds it: across every constant swept, the mean absolute score change is at most 1.15 points, the maximum below three, and the minimum Spearman correlation with the baseline is 0.9924. The interpretation is not that the constants are correct, which the data cannot show, but that the architecture is not balanced on a knife edge: a necessary condition for the design to be worth validating, not a substitute for it.

TABLE III. SCORE STABILITY UNDER CONSTANT SWEEPS

Constant	Range swept (nominal)	mean ΔS	max ΔS	min ρ
Pitch scale θp	90-150 Hz (120)	0.95	2.78	0.9924
HNR scale θh	22-38 dB (30)	1.15	2.97	0.9928
Hedge divisor	5-12 (8)	0.86	2.50	0.9955
Filler divisor	8-18 (12)	0.83	2.23	0.9974
Pause divisor	4-10 (6)	0.71	2.67	0.9960
Speech-rate band	110-170 / 130-150 WPM	0.38-0.39	2.34	0.9926

D. Robustness to Correlated Degradation

The preceding experiments sample inputs independently, understating a realistic failure mode: one poor microphone or camera degrades several signals together. We repeated Experiment 1 on a second population of 20,000 vectors generated from two latent factors, recording quality and speaker fluency, each observable loading on its factor plus private noise. This induces substantial dependence, for example +0.71 between HNR and facial confidence, and a wider score distribution (standard deviation 13.40 against 7.36). The conclusion strengthens rather than weakens: pooled mean absolute disturbance is 2.82 points for renormalization against 9.09 for zero-imputation, a ratio of 3.23 against 3.00 under independent sampling, and grade-band disturbance falls to 26.3% against 62.6%. Correlated loss is the case renormalization is designed for: when a modality degrades, the surviving families still carry evidence, whereas zero-imputation charges the candidate for every signal the equipment failed to capture.

E. Invariance and Behavioural Tests

The simulation characterizes the scoring function; the test suite verifies that the implementation enforces the boundary the function assumes. Two invariance tests assert that varying LLM-generated evaluation scores, and varying diagnostic aggregates, leave the official score unchanged. Further tests cover grade-boundary correctness, Mode A and Mode B behaviour, fusion-preview weight consistency, the face-required gate, the no-scorable-family case, and the distinction between a genuinely silent candidate and a failed transcription pipeline: both produce a zero word count while warranting opposite treatment, one being evidence about the candidate and the other about the recording.

VIII. DISCUSSION

The results support a narrow but concrete claim: for a deterministic multimodal assessment score, the treatment of missing data is a design decision with a measurable cost of roughly three times in score disturbance and more than twice in grade-band disturbance. That cost falls on candidates whose recordings were defective, close to the worst incidence for an assessment instrument, since equipment quality tracks circumstance rather than competence.

The speech-emotion case illustrates why the exclusion boundary earns its cost. The available recognizer is trained on acted emotional speech and produced highly confident negative-emotion predictions on calm academic presentations; admitting it because it was computable would have penalized well-composed candidates for a corpus mismatch. The same reasoning governs gaze events, head turn and tilt, response structure, energy consistency and the temporal CNN-LSTM model. The sensitivity result nonetheless complicates that principle: the boundary keeps unvalidated signals out but cannot make the admitted signals valid, and the two admitted signals with the weakest external validation carry the most weight. We take this as the central limitation of the present contribution.

IX. THREATS TO VALIDITY

A. Construct Validity and Simulation Assumptions

The largest threat is construct validity. Stage-1 measures behavioural and delivery characteristics, not academic competence: a candidate may have fluent delivery and weak understanding, or the reverse. The score should be read as an automated delivery measure supporting an examiner, and the equal-weight Mode B fusion is not a claim that delivery and content deserve equal weight, an institutional policy decision. The simulation samples uniformly over engineering-plausible ranges rather than from an observed cohort, so absolute score levels are not predictions about real candidates. The independence assumption is tested in Section VII-D, where correlated sampling widened rather than narrowed the reported advantage.

B. Ground Truth, Fairness and Feature-Level Validity

No independent examiner-scored dataset is available, so the paper establishes neither agreement with human assessors, nor fairness across subgroups, nor predictive validity, nor generalization across institutions, accents, cameras or lighting. Accent is a particular concern: Whisper's word error rate varies across accents and transcript-derived features inherit that variation, so a transcription-quality effect could present as a delivery-quality difference. Several features remain heuristic, including gaze and head-event detection, the untrained CNN-LSTM model and manually-tuned head-pose thresholds, all recorded in the ledger.

C. Scope Relative to the Original Project Plan

The university-approved proposal envisaged a broader system including semantic content grading, an explainable-AI

layer, a second-marking mechanism and libraries including DeepFace. The component reported here is narrower: technical accuracy is an optional external Mode-B input, formal explainability and second marking are not implemented, and the system uses MediaPipe with a custom EfficientNet-B0 emotion model. We report the system as built rather than as planned, because reproducibility depends on the description matching the artefact.

X. The Required Validation Study and Future Work

Because the absence of human-agreement evidence is the principal limitation of this work, we specify the study needed to remove it, registered here in advance of data collection. *Materials:* at least 20 recorded viva sessions from an authentic examination setting, under informed consent and ethics approval, with device, room, lighting and microphone logged so recording condition enters as a covariate rather than a confound. *Raters:* each session scored independently by at least three qualified examiners on sub-scales for communication, presentation, engagement and technical content, using a written rubric, blind to the system output and to each other, with rating order counterbalanced against fatigue [1].

Analyses: inter-rater reliability among examiners is computed first and reported regardless of outcome, since it sets the ceiling against which system-human agreement must be read, with Krippendorff's alpha suiting ordinal sub-scales and more than two raters. Agreement then follows as Spearman correlation and mean absolute error against the mean human sub-scale score, with bootstrap intervals and a grade-band confusion matrix. Sub-scale comparison matters more than a single aggregate: Stage-1 claims to measure delivery, so correlation with human *communication* ratings is the direct test, whereas high correlation with *technical* ratings would indicate construct contamination. Secondary analyses cover ablation of the engagement family, subgroup analysis by language background, and a degradation study on real recordings.

Poor agreement among examiners would be a result about the assessment task, not a licence to prefer the system's numbers; weak correlation would indicate the behavioural construct is insufficient for grading. Neither outcome would justify autonomous grading. Beyond validation, the priorities are testing held-out signals against human labels, replacing step-valued normalizers with data-calibrated functions, and reference-grounded semantic grading with attribution using SHAP [15].

XI. Conclusion

This paper presented Stage-1 v1.1, a deterministic multimodal scoring architecture for automated viva assessment fusing visual engagement, acoustic delivery and transcript fluency under an explicit missing-data policy and an enforced exclusion boundary. Measured over 20,000 sampled feature vectors and driven by the production scoring code itself, renormalizing weights away from unmeasurable signals reduces mean absolute score disturbance by a factor of 3.00 relative to zero-imputation, eliminates a downward bias reaching 22.6 points when a whole modality is lost, and more than halves grade-band disturbance. No single input moves the score by more than 1.52 points under a 10% perturbation, and ordering is stable across every unfitted constant (rho >= 0.992). The advantage widens to 3.23 when degradation is correlated across signals, the realistic failure mode. The analysis also surfaced a limitation the architecture cannot resolve alone: the two highest-leverage inputs are the two with the weakest external validation. The system therefore offers a reproducible and auditable foundation rather than a validated grader, and the human-examiner study in Section X is the necessary next step before any deployment affecting a student's mark.

Acknowledgment

The authors thank Ms. Sanjeewi Chandrasiri, SLIIT, for supervising the broader research project.

References

[1] A. Gurung, A. F. Botelho, R. Thompson, A. C. Sales, S. Baral, and N. T. Heffernan, “Considerate, unfair, or just fatigued? Examining factors that impact teacher grading,” in Proc. 30th Int. Conf. Computers in Education (ICCE), 2022.

[2] Y. Tao, E. Coltey, T. Wang, M. Alonso, M.-L. Shyu, S.-C. Chen, H. Alhaffar, A. Elias, B. Bogosian, and S. Vassigh, “Confidence estimation using machine learning in immersive learning environments,” in Proc. IEEE 3rd Int. Conf. Multimedia Information Processing and Retrieval (MIPR), 2020, pp. 247–252, doi: 10.1109/MIPR49039.2020.00058.

[3] K. N. Reddy, K. Medikonda, R. N. Sree, and P. B. Pati, “Understanding the student confidence in viva responses through audio analysis,” in Proc. 3rd Int. Conf. Integrated Circuits and Communication Systems (ICICACS), 2025, pp. 73–80.

[4] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, “Multimodal machine learning: A survey and taxonomy,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 41, no. 2, pp. 423–443, 2019, doi: 10.1109/TPAMI.2018.2798607.

[5] M. Wang, S. Fan, Y. Li, Z. Xie, and H. Chen, “Missing-modality enabled multi-modal fusion architecture for medical data,” J. Biomed. Inform., vol. 164, art. 104796, 2025, doi: 10.1016/j.jbi.2025.104796.

[6] R. A. Khalil, E. Jones, M. I. Babar, T. Jan, M. H. Zafar, and T. Alhussain, “Speech emotion recognition using deep learning techniques: A review,” IEEE Access, vol. 7, pp. 117327–117345, 2019, doi: 10.1109/ACCESS.2019.2936124.

[7] C. Lugaresi et al., “MediaPipe: A framework for building perception pipelines,” 2019, arXiv:1906.08172. [Online]. Available: https://arxiv.org/abs/1906.08172

[8] X. Ben, Y. Ren, J. Zhang, S.-J. Wang, K. Kpalma, W. Meng, and Y.-J. Liu, “Video-based facial micro-expression analysis: A survey of datasets, features and algorithms,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 44, no. 9, pp. 5826–5846, 2022, doi: 10.1109/TPAMI.2021.3067464.

[9] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in Proc. 40th Int. Conf. Machine Learning, PMLR, vol. 202, 2023, pp. 28492–28518.

[10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in Proc. NAACL-HLT, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

[11] P. Senarathne, M. Silva, A. Methmini, D. Kavinda, and S. Thelijjagoda, “Automate traditional interviewing process using natural language processing and machine learning,” in Proc. 6th Int. Conf. for Convergence in Technology (I2CT), 2021, pp. 1–6, doi: 10.1109/I2CT51068.2021.9418115.

[12] H. Cao and S. Zahid, “Automated viva voce using generative AI for student coursework authentication,” in Proc. Int. Conf. Educational Technology and Artificial Intelligence (ETAIC), ACM, 2025, doi: 10.1145/3766557.3766564.

[13] P. Ipeirotis and K. Rizakos, “Scalable and personalized oral assessments using voice AI,” Commun. ACM, 2026, doi: 10.1145/3831714.

[14] M. T. Ribeiro, S. Singh, and C. Guestrin, “‘Why should I trust you?’: Explaining the predictions of any classifier,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.

[15] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 4765–4774.